

Pythagoras' Theorem & Trigonometry

Extra practice test 4 · 27 marks total

Name: _____

Date: _____

Score: _____ / 27

Section A — Pythagoras' theorem

A1. A right-angled triangle has shorter sides of 10 cm and 24 cm. Find the length of the hypotenuse.

[2 marks]

A2. A right-angled triangle has a hypotenuse of 20 cm and one shorter side of 12 cm. Find the length of the other shorter side.

[3 marks]

A3. A rectangle measures 18 cm by 24 cm. Find the length of its diagonal.

[2 marks]

Section B — SOH CAH TOA basics

B1. In a right-angled triangle, what do we call the side next to angle θ that is not the hypotenuse?

[1 mark]

B2. Write down the SOH CAH TOA formula you would use to find the opposite side if you know the adjacent side and the angle.

[1 mark]

B3. Write down the SOH CAH TOA formula you would use to find an angle if you know the adjacent and hypotenuse.

[1 mark]

Section C — Finding missing sides

C1. A right-angled triangle has an angle of 33° and a hypotenuse of 16 cm. Find the length of the side opposite the angle, to 1 decimal place.

[3 marks]

C2. A right-angled triangle has an angle of 47° and an adjacent side of 6 cm. Find the length of the hypotenuse, to 1 decimal place.

[2 marks]

C3. A right-angled triangle has an angle of 58° and an adjacent side of 3 cm. Find the length of the side opposite the angle, to 1 decimal place.

[2 marks]

Section D — Finding missing angles

D1. A right-angled triangle has an opposite side of 9 cm and a hypotenuse of 15 cm. Find the angle, to 1 decimal place.

[2 marks]

D2. A right-angled triangle has an adjacent side of 6 cm and an opposite side of 13 cm. Find the angle, to 1 decimal place.

[3 marks]

Section E — Applying Pythagoras & trigonometry

E1. A ladder rests against a vertical wall. The foot of the ladder is 2.4 m from the wall and the ladder is 6.5 m long. Calculate how far up the wall the ladder reaches, to 1 decimal place.

[2 marks]

E2. A ramp rises at an angle of 15° to the horizontal over a horizontal distance of 12 m. Calculate the height gained by the ramp, to 1 decimal place.

[3 marks]